The Vacuus of the Queens

the Mirror Law: which faces of a reflection polytope lie on its mirrors, read from its Coxeter chain; the 600-cell and the 120-cell share the 60 mirrors of their group, and the bitruncated middle carries both Queens' faces on every mirror

DICTO (artist and geometer, Osaka, Japan) · dictospheres.com · contact@dictospheres.com · ORCID 0009-0001-2620-2036
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Take a solid and look for its *Vacuus* (#87): the flat slices through its centre that carry its faces. The octahedron's twelve edges lie on three squares through its centre; the cube's edges lie on no such slice at all. In four dimensions the same question asks which 2-dimensional faces lie in a 3-dimensional slice through the centre. This paper answers it for the Queens, the 600-cell (the White Queen) and the 120-cell (the Red Queen) of #63, #65 and #76, for the stages between them, and for every polytope built from a reflection group, by one rule read off the group's Coxeter chain.

1. The setting

A finite reflection group in n dimensions is generated by reflections in n mirrors whose unit normals $\alpha_1, \ldots, \alpha_n$ are its simple roots; the mirrors of the group are the hyperplanes orthogonal to its roots, and every root is a combination of simple roots whose coefficients are all ≥ 0 or all ≤ 0 (Humphreys 1990, §1.3). Its Coxeter diagram joins nodes i, j whose mirrors meet at angle π/m, $m \geq 3$, by a bond labelled m. The groups here have diagrams that are chains: B_3 4–3 (the cube), H_3 5–3 (the icosahedron), A_4 3–3–3 (the 5-cell), B_4 4–3–3 (the tesseract), F_4 3–4–3 (the 24-cell) and H_4 5–3–3 (the Queens).

Wythoff's construction places a seed point x with labels $\lambda_i = \alpha_i \cdot x \geq 0$ and takes the hull of its images. The nodes with $\lambda_i > 0$ are the *ringed* nodes. Its faces are the images of the faces $F_J = \text{hull}(W_J\, x)$, where W_J is generated by the reflections of a node set J whose every connected piece contains a ringed node (Coxeter 1973; Champagne, Kjiri, Patera and Sharp 1995). The orbit study of 11 November read these seeds back from the corners of DICTO's solids: the Queens' path is the seed sliding along the H_4 chain, from the 120-cell (1, 0, 0, 0) through the truncated 120-cell (1, 1, 0, 0), the bitruncated 600-cell (0, 1, 1, 0) and the truncated 600-cell (0, 0, 1, 1) to the 600-cell (0, 0, 0, 1); the golden loop of DICTONIC is the B_3 seed (0, x, 1).

2. The Mirror Law

Theorem 1 (the Mirror Law). Let P be the Wythoff polytope of a seed x, and J a node set of size $n - 2$ giving faces of P (so they have dimension $n - 2$: edges in three dimensions, 2-faces in four). The faces of type J lie on mirrors of the group if and only if the roots of the *unringed* nodes include one orthogonal to α_j for every j in J.

Proof. The face F_J lies in the affine space x + span(α_J), so the hyperplane through the centre that contains it is span(x, α_J), and its other faces of type J are images of F_J, which carry mirrors to mirrors. That hyperplane is a mirror exactly when its normal is a root, that is, when some root r is orthogonal to x and to every α_j, $j \in$ J. Write $r = \Sigma\, c_l\, \alpha_l$, all c_l of one sign. Then $r \cdot x = \Sigma\, c_l\, (\alpha_l \cdot x)$ after expressing x in the basis dual to the simple roots, which is $\Sigma\, \lambda_l\, c_l$; with $\lambda_l > 0$ on the ringed nodes and every c_l of one sign, this vanishes exactly when $c_l = 0$ on every ringed node, so that r is a root of the unringed nodes. ∎

The law was checked against direct measurement, the normal of span(x, α_J) compared with every root, for every seed of zeros and ones and every face type in A_3, B_3, H_3, A_4, B_4, F_4 and H_4: 228 cases, no disagreement (mirror_law.py).

Corollary 2 (the Queens). Every face of a regular polytope lies on a mirror. The 1200 triangles of the 600-cell and the 720 pentagons of the 120-cell lie on the same 60 central slices, the 60 mirrors of H_4: 20 triangles on each, and 12 pentagons on each.

Proof. A regular polytope has one ringed node at an end of the chain, node 1 say, and one face type J = {1, 2} (in four dimensions); node 4 is unringed and α_4 is orthogonal to α_1 and α_2. So every face lies on a mirror. Every mirror carries the same number of faces because the group moves any mirror to any other (its roots form one orbit), so 1200 / 60 = 20 and 720 / 60 = 12; that no other slice carries a face, and that both Queens use all 60 mirrors, was checked by computing the central hyperplane of every face (mirror_check.py). ∎

Corollary 3 (the middles). For a four-node chain 1–2–3–4 and the seed (0, 1, 1, 0), the faces of the two end bonds lie on mirrors and the faces of the middle bond (2m-gons, m its label) lie on none. In B_4 and H_4 these are exactly the faces between cells of different kinds.

Proof. The face types are {1, 2}, {2, 3} and {3, 4}; the unringed nodes 1 and 4 are not joined, so their only roots are $\pm\alpha_1$ and $\pm\alpha_4$. α_4 is orthogonal to α_1 and α_2, and α_1 to α_3 and α_4, so the end faces lie on mirrors (Theorem 1); neither is orthogonal to both α_2 and α_3, since node 1 is joined to 2 and node 4 to 3, so the middle faces lie on none. A middle face lies between a cell of type {1, 2, 3} and one of type {2, 3, 4}; they are congruent when the chain reads the same from both ends (A_4 3–3–3, F_4 3–4–3) and differ otherwise. ∎

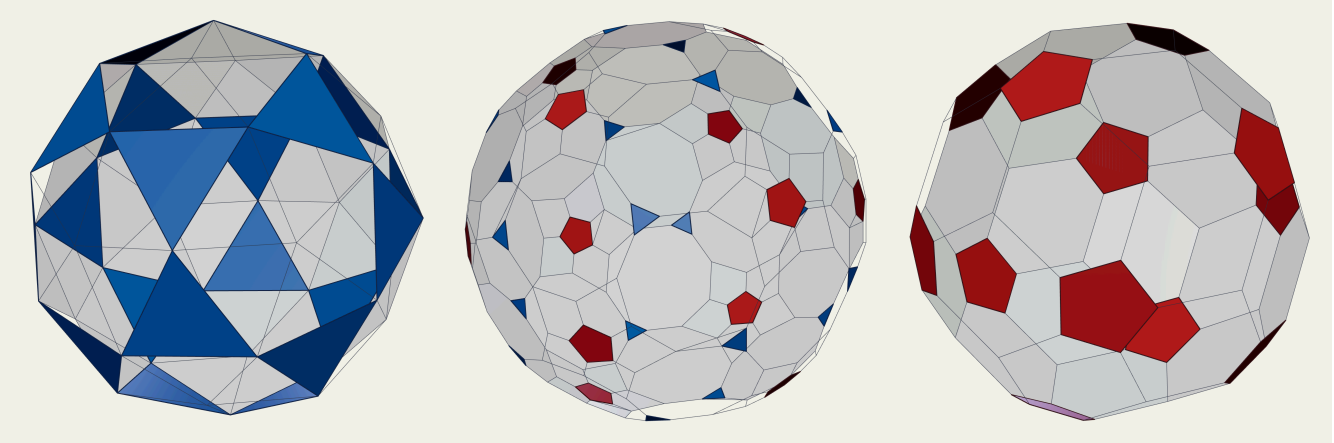


Figure 1. One mirror of H_4 cutting each Queen and their middle; each slice is drawn as a 3D solid in grey glass, its back half seen through. Left, the 600-cell (the slice has 42 corners and 80 triangles): the mirror carries 20 of the White Queen's triangles (blue). Right, the 120-cell (80 corners): 12 of the Red Queen's pentagons (red). Middle, the bitruncated 600-cell (240 corners): the same mirror carries 20 triangles and 12 pentagons, the faces between its truncated tetrahedra and between its truncated icosahedra; none of its hexagons, where the two kinds meet.

3. The five stages between the Queens

stage (seed)	2-faces	on the 60 mirrors, on each	on no mirror
600-cell (0, 0, 0, 1)	1200 triangles	20 triangles	none
truncated 600-cell (0, 0, 1, 1)	2400 triangles, 1200 hexagons	20 hexagons	the 2400 triangles
bitruncated 600-cell (0, 1, 1, 0)	1200 triangles, 720 pentagons, 2400 hexagons	20 triangles and 12 pentagons	the 2400 hexagons
truncated 120-cell (1, 1, 0, 0)	2400 triangles, 720 decagons	12 decagons	the 2400 triangles
120-cell (1, 0, 0, 0)	720 pentagons	12 pentagons	none

Each row follows from Theorem 1 (for the truncated 600-cell, the unringed nodes 1 and 2 give the roots of the pentagon group, and α_1 is orthogonal to the hexagons' nodes 3, 4 but no root of nodes 1, 2 is orthogonal to the triangles' nodes 2, 3) and was measured face by face (vacuus_queens.py: every face's central hyperplane, grouped and classed under all 14 400 symmetries). The faces on no mirror are scattered: computed, they lie two on each of 1200 central slices. At the bitruncated 600-cell the hexagons on no mirror are the faces where a truncated tetrahedron meets a truncated icosahedron: the faces of the exchanges in #104's scutoid tissue. The faces between like cells stand on the mirrors; the faces of exchange stand off them.

4. Every chain's middle

chain	middle (0, 1, 1, 0)	on the mirrors	on no mirror (middle bond)
A_4 3–3–3, 10 mirrors	bitruncated 5-cell: 30 corners, 10 truncated tetrahedra	20 triangles	20 hexagons
B_4 4–3–3, 16 mirrors	bitruncated tesseract: 96 corners, 8 truncated octahedra, 16 truncated tetrahedra	32 triangles (between truncated tetrahedra), 24 squares (between truncated octahedra)	64 hexagons, between the two kinds
F_4 3–4–3, 24 mirrors	bitruncated 24-cell: 288 corners, 48 truncated cubes	192 triangles	144 octagons
H_4 5–3–3, 60 mirrors	bitruncated 600-cell: 3600 corners, 120 truncated icosahedra, 600 truncated tetrahedra	1200 triangles, 720 pentagons	2400 hexagons, between the two kinds

Each middle was built from its seed and its faces measured (vacuus_middles.py; the H_4 row repeats Section 3 as a control). In the self-reading chains A_4 and F_4 every cell is alike, yet the middle-bond faces still leave the mirrors: the law belongs to the bond, not to the cells.

5. Back in three dimensions

Corollary 4. For the golden loop T(x), the B_3 seed (0, x, 1), the 12 edges of type {3} lie on the three coordinate mirrors and the 24 edges of type {2} lie on none.

Proof. Node 1 is the only unringed node; α_1 is orthogonal to α_3 (nodes 1 and 3 are not joined) but not to α_2. ∎ This is the split seen in the 3D Vacuus of the DICTO 1:2 Truncated Octahedron (#105, Figure 4) and measured on the Icosaxes and the Unborn Axes: three discs of four edges, the other edges each on a central plane of its own with one partner.

6. Sources and sources checked

DICTO: #63, #65, #76 (the Queens and their stages); #87 (the Vacuus); #104 Scutoid Tissues in Four Dimensions (the exchanges at the bitruncated 600-cell); #105 DICTO 1:2 Truncated Octahedron: The 4D Vacuus of the Permutohedron (Figure 4). H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover (1973); B. Champagne, M. Kjiri, J. Patera and R. T. Sharp, Description of reflection-generated polytopes using decorated Coxeter diagrams, Can. J. Phys. 73 (1995) 566–584; J. E. Humphreys, *Reflection Groups and Coxeter Groups*, Cambridge University Press (1990); N. Whitehead, Classifying regular polyhedra and polytopes using Wythoff's construction, arXiv:2001.09364. Checked for a prior statement of Theorem 1: DICTO's reference library (its cited, bridges and expansion lists; entries on Wythoff's construction and root systems: Moody and Patera 1993 only), and a web search for faces of Wythoff polytopes lying in reflection hyperplanes (results: Whitehead 2020 and general pages on uniform polytopes and Wythoff symbols; Champagne et al. 1995 named there, not read in full). Nothing found; the proof is short and may be known.